power

Power: battery input, 5 V bucks, SoM eFuse, 5 V port switches, 3V3 / 1V8 / VCCO

som

SoM connectors JX1–JX4 (BergStak 0.8 mm), test points

fc

Flight controller: IMU + heater, ADS1115, RGB LED, buzzer driver, status LEDs

motor

PWM 1–8 and 4–in–1 ESC: DIR buffers, BSS138 clamps

bus

CAN1/2, external I2C, external SPI, RF–CTL, 5 V AUX

conn

USB–C OTG, Ethernet 100M, microSD, debug (console + JTAG), reset

2–6S in, IMU, Pixhawk DS–009 ports, ESC, PWM 1–8, CAN, USB–C OTG, ETH 100M, microSD

NyxHop

Sheet: /

File: nyxhop_baseboard.kicad_sch

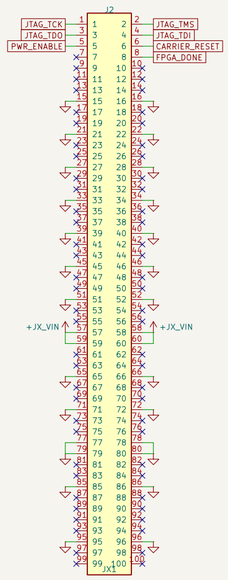
Title: NyxHop air–unit baseboard v0 (carrier for ADRV9364–Z7020 / ADRV93

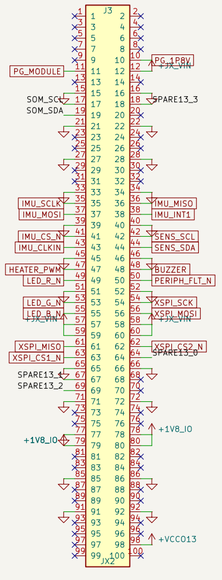
Size: A4	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 1/8

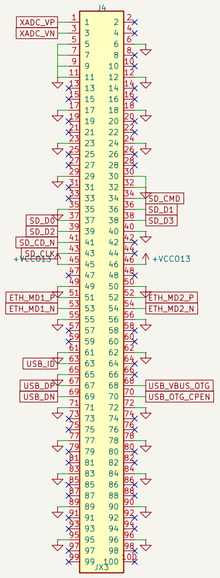


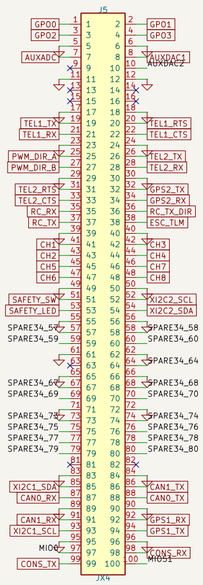
Power: battery input, 5 V bucks, SoM eFuse, 5 V port switches, 3V3 / 1V8 / VCCO

NyxHop		
Sheet: /power/		
File: power.kicad_sch		
Title: NyxHop baseboard v0 — Power		
Size: A1	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 2/8









TP16 SOM_SCL
TP17 SPARE13_3
TP18 SOM_SDA
TP19 SPARE13_0
TP20 SPARE13_1
TP21 SPARE13_2
TP22 AUXDAC2
TP23 SPARE34_57
TP24 SPARE34_58

TP25 SPARE34_59
TP26 SPARE34_60
TP27 SPARE34_64
TP28 SPARE34_67
TP29 SPARE34_68
TP30 SPARE34_69
TP31 SPARE34_70
TP32 SPARE34_73
TP33 SPARE34_74
TP34 SPARE34_75
TP35 SPARE34_76
TP36 SPARE34_77
TP37 SPARE34_78
TP38 SPARE34_79
TP39 SPARE34_80
TP40 MIO0
TP41 MIO51

H1 M3
H2 M3
H3 M3
H4 M3

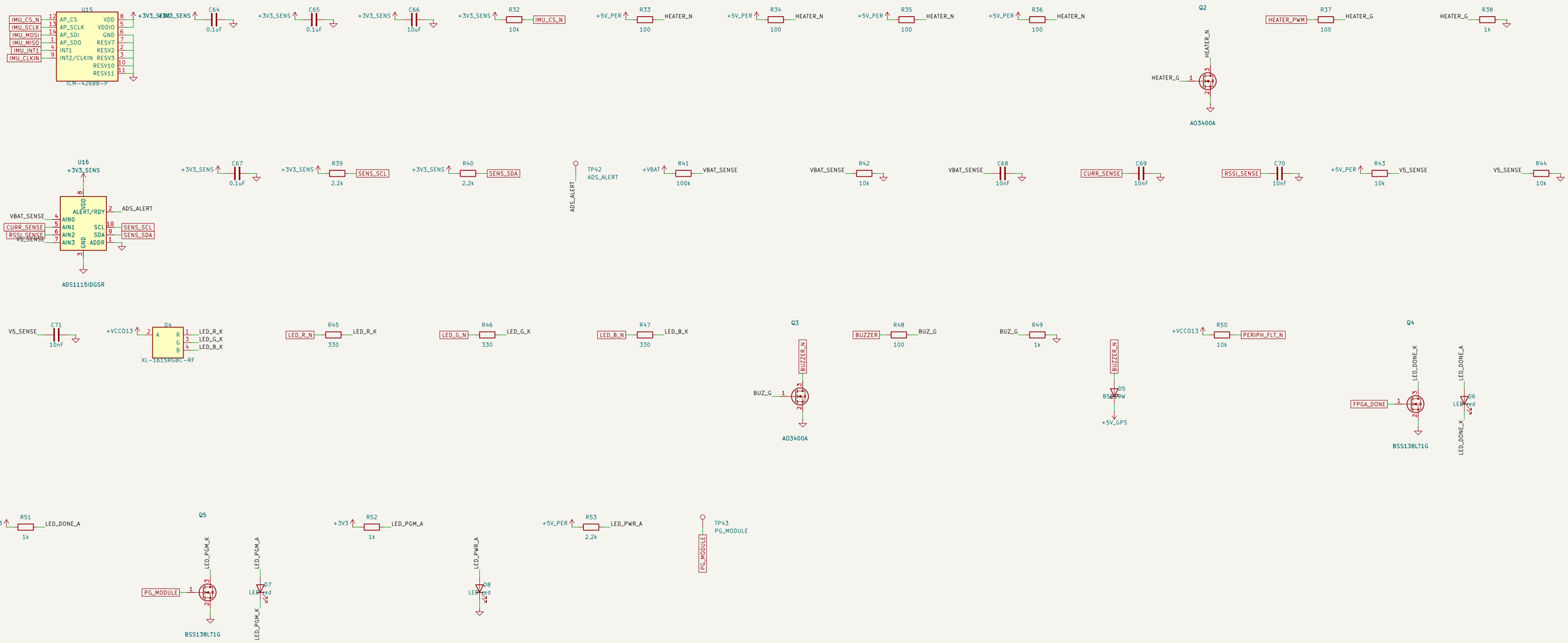
SoM connectors JX1–JX4 (BergStak 0.8 mm), test points

NyxHop

Sheet: /som/
File: som.kicad_sch

Title: NyxHop baseboard v0 – SoM connectors JX1–JX4 (BergStak 0.8 mm),

Size: A2	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 3/8



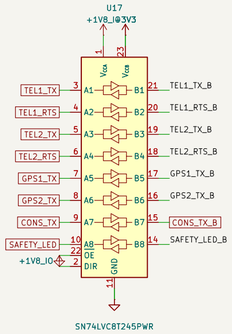
Flight controller: IMU + heater, ADS1115, RGB LED, buzzer driver, status LEDs

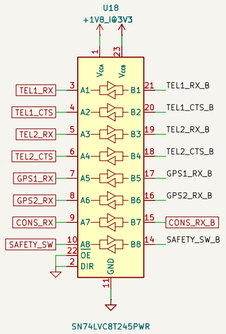
NyxHop

Sheet: /fc/
File: fc.kicad_sch

Title: NyxHop baseboard v0 — Flight controller

Size: A2	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 4/8

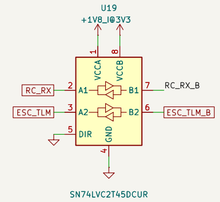


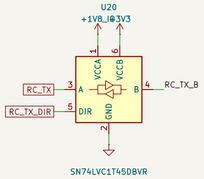




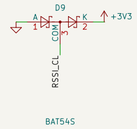






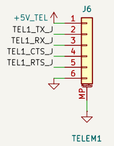


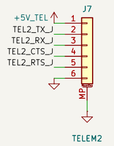


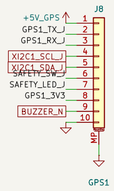


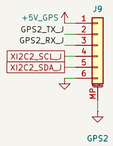


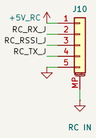


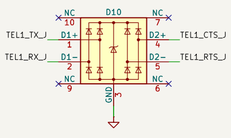


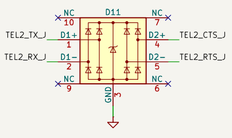


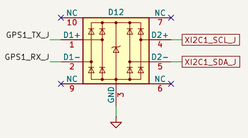


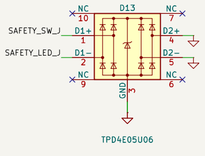


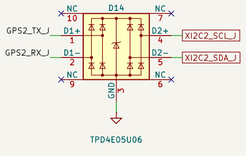


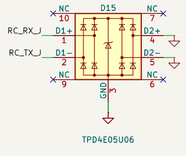












TELEM1/2, GPS1/2, RC IN, console: level shift 1.8 <–> 3.3 V

NyxHop

Sheet: /uart/
File: uart.kicad_sch

Title: NyxHop baseboard v0 – TELEM1/2, GPS1/2, RC IN, console

Size: A2	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 5/8

U21
+1V8_IO
+3V3
VCCA
VCCB
CH1
CH2
PWM_DIR_A
A1
A2
DIR
GND
B1
B2
CH1_B
CH2_B
SN74LVC2T45DCUR
C80
0.1uF
C81
U22
CH3
CH4
CH3_B
CH4_B
C82
C83
U23
CH5
CH6
PWM_DIR_B
CH5_B
CH6_B
C84
C85
U24
CH7
CH8
CH7_B
CH8_B
C86
C87
R83
PWM_DIR_B
1k
R84
+5V_PER
CLAMP_G
10k
C88
0.1uF
R85
CH1_B
CH1_P
47
Q6
CH1_PJ
BSS138LT1G
R86
100k
D16
PESD24VS2UL
R87
CH2_B
CH2_P
Q7
CH2_PJ
D17
R88
R89
CH3_B
CH3_P
Q8
CH3_PJ
R90
CH3_PJ
D18
R91
CH4_B
CH4_P
Q9
CH4_PJ
R92
D19
R93
CH5_B
CH5_P
Q10
CH5_PJ
R94
D20
R95
CH6_B
CH6_P
Q11
CH6_PJ
R96
D21
R97
CH7_B
CH7_P
Q12
R98
D22
R99
CH8_B
CH8_P
Q13
CH8_PJ
R100
D23
R101
CH1_B
CH1_E
Q14
M1_J
R102
D24
R103
CH2_B
CH2_E
Q15
M2_J
R104
M2_J
D25
R105
CH3_B
CH3_E
Q16
M3_J
R106
D26
R107
CH4_B
CH4_E
Q17
M4_J
R108
D27
Q18
TLM_J
R109
TLM_P
ESC_TLM_B
R110
D28
D29
BAT54S
CURR_CL
+3V3
R111
ESC_CURR_J
10k
R112
47k
CURR_SENSE
R113
ESC_VBAT_J
+VBAT
0
J11
M1
ESC 4in1
J12
M2
PWM 1-8
TP44
PWM_VSERVO

PWM 1-8 and 4-in-1 ESC: DIR buffers, BSS138 clamps

NyxHop

Sheet: /motor/

File: motor.kicad_sch

Title: NyxHop baseboard v0 – PWM 1-8 and 4-in-1 ESC

Size: A2

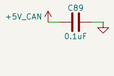
Date:

Rev: 0.2

KiCad E.D.A. 10.0.6

Id: 6/8

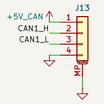


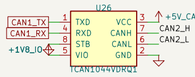


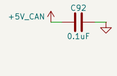






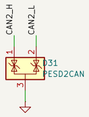


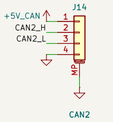


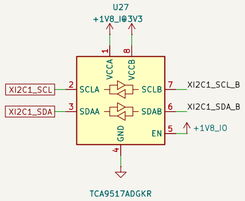






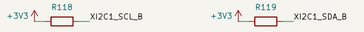


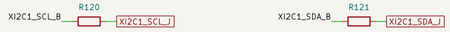


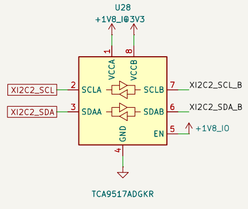












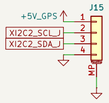


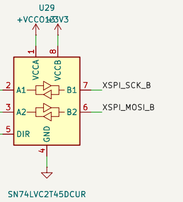


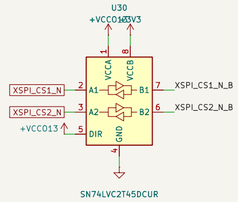


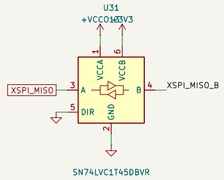






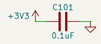


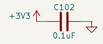


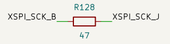
















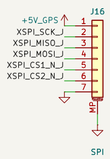


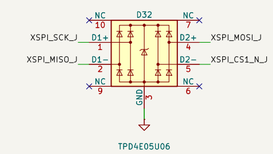


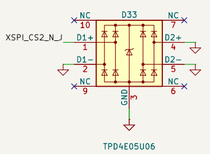




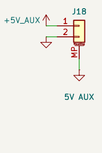












CAN1/2, external I2C, external SPI, RF-CTL, 5 V AUX

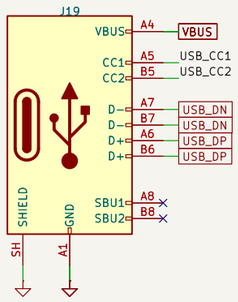
NyxHop

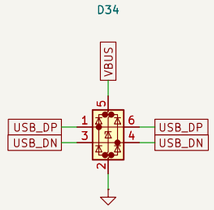
Sheet: /bus/

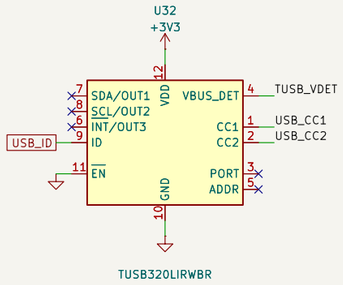
File: bus.kicad_sch

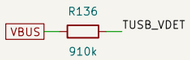
Title: NyxHop baseboard v0 – CAN1/2, external I2C, external SPI, RF-CTL, 5

Size: A2	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 7/8



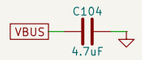








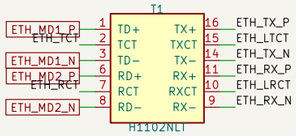












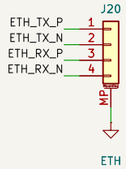


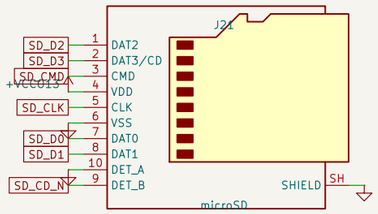


















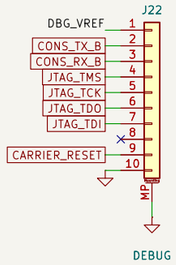
















USB-C OTG, Ethernet 100M, microSD, debug (console + JTAG), reset

NyxHop

Sheet: /conn/
File: conn.kicad_sch

Title: NyxHop baseboard v0 — USB-C OTG, Ethernet 100M, microSD, debug (

Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 8/8